W I N M O

Leveraged ETF Protocol

Non-Liquidatable 2x Long & -1x Inverse
Leveraged Exposure to RWA & Crypto

Whitepaper v1.1 | March 2026

65 ERC-20 Tokens · 5 Asset Classes · Hyperliquid Execution

Stocks · Indices · Commodities · Forex · Crypto

Winmo Foundation

EXECUTIVE SUMMARY

THE PROBLEM

The leveraged ETF market exceeds $60B in TradFi (ProShares TQQQ, Direxion SPXL). In DeFi, no product provides permissionless, non-liquidatable leveraged exposure to real-world asset prices or crypto. IndexCoop's ETH2x-FLI was sunset due to structural costs. There is zero on-chain access to leveraged S&P 500, Apple, Gold, or EUR/USD.

THE SOLUTION

Winmo creates ERC-20 tokens that provide daily 2x leveraged (long) and -1x inverse (short) exposure to stocks, indices, commodities, forex, and crypto. Users deposit USDC and receive a token (e.g., sAAPL-2x or sBTC-1xS). A four-layer rebalancing system ensures the product cannot be liquidated. All positions execute on Hyperliquid.

65 ERC-20 Tokens	**5** Asset Classes	**$0** Liquidation Risk
2x / -1x Long & Inverse	**24/7** On-Chain Trading	**$3.4M** Revenue at $100M AUM

HOW IT WORKS

1. User deposits USDC into a vault and receives an ERC-20 token (e.g., sBTC-2x).
2. The vault opens a 2x leveraged perpetual futures position on Hyperliquid.
3. Daily rebalancing resets leverage to 2.0x. Four safety layers prevent liquidation.

COMPETITIVE ADVANTAGE

Feature	ProShares TQQQ	IndexCoop FLI	Winmo
Assets	QQQ only	ETH only	65 (RWA + Crypto)
Directions	Long OR Inverse	Long only	Long AND Inverse
Permissionless	No (KYC)	Yes	Yes
24/7 Trading	No	Yes	Yes
Non-liquidatable	N/A (TradFi)	No (Aave liq.)	Yes (4-layer)
Mechanism	Swaps/futures	Lending (Aave)	Perps (Hyperliquid)

Keywords: Leveraged ETF, Inverse ETF, Non-Liquidatable, DeFi, Perpetual Futures, ERC-4626, Hyperliquid, Volatility Decay

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I. THE OPPORTUNITY

A. The Product Gap

Traditional leveraged ETFs (ProShares TQQQ, Direxion SPXL) provide 2-3x daily leveraged exposure to indices and are among the most traded products in TradFi. The combined AUM of leveraged and inverse ETFs exceeds $60 billion. Yet in DeFi, this product category barely exists.

IndexCoop created the ETH2x-FLI for Ethereum exposure using Aave/Compound lending. It was ultimately sunset due to high costs, limited asset coverage, and low capital efficiency. No product exists today that provides permissionless, 24/7, non-liquidatable leveraged exposure to real-world asset prices like Apple stock, the S&P 500, or Gold - let alone both long and inverse.

B. Why Perp-Based > Lending-Based

IndexCoop's FLI used Aave lending for leverage (deposit ETH, borrow USDC, buy more ETH). This approach has structural limitations that Winmo's perp-based model eliminates:

Dimension	Lending (FLI)	Perps (Winmo)
Capital efficiency	Low (overcollateralized)	High (direct leverage)
Asset coverage	ETH, BTC only	Any asset with perp market
Rebalancing cost	High (swap + repay)	Low (single position adjust)
Gas cost	Very high (L1)	Low (Arbitrum / HL L1)
Outcome	Sunset	Active design

C. Why Now

Hyperliquid now supports both crypto and RWA perpetual futures (stocks, indices, commodities, forex) with deep liquidity, sub-second price updates, and low fees. This makes it possible for the first time to build a permissionless leveraged ETF protocol covering all major asset classes from a single execution layer.

> **KEY TAKEAWAY**
>
> Winmo is the first permissionless protocol offering both 2x long AND -1x inverse leveraged exposure to stocks, commodities, indices, forex, and crypto - all non-liquidatable, 24/7, on Hyperliquid. This product category does not yet exist in DeFi.

II. LESSONS FROM INDEXCOOP

A. What IndexCoop Built

IndexCoop launched ETH2x-FLI and BTC2x-FLI in 2021 - the first DeFi leveraged tokens. They used a lending loop on Aave: deposit ETH as collateral, borrow USDC, swap USDC for more ETH, redeposit, and repeat until ~2x exposure is achieved. The products were ERC-20 tokens built on Set Protocol, peaked at ~$200M combined AUM, and were ultimately sunset in 2023.

B. Why FLI Failed:Structural Autopsy

FLI's failure was architectural, not operational. The lending-based mechanism has inherent limitations that cannot be fixed with better engineering:

Failure Mode	Root Cause	Impact
Gas costs ($100-500/rebal)	3-5 L1 txns per rebalance (supply + borrow + swap)	Costs absorbed by NAV, eroding returns
Aave parameter dependency	Aave governance controls LTV, rates, freezes	FLI had no control over its own leverage engine

Only ETH & BTC	Requires deep Aave lending market per asset	No stocks, gold, forex, or altcoins possible
No inverse products	Lending loop can only go long	Entire product category structurally impossible
Leverage drift (1.7x-2.3x)	Rebalancing delayed to save gas	Imprecise exposure, tracking error
Aave liquidation risk	Flash crash = collateral ratio drops	May 2021 crash nearly liquidated FLI
Swap slippage on Uniswap	Large rebalances = price impact on spot DEX	$10M+ rebalances moved markets against FLI
Borrow rate volatility	Aave rates spiked to 90%+ in DeFi summer	Unpredictable, sometimes extreme leverage cost

C. How Winmo Addresses Every FLI Failure

FLI Problem	Winmo Solution
$100-500 gas per rebalance	$0.01-0.10 on Arbitrum / Hyperliquid L1
Aave dependency (parameters, freezes)	Hyperliquid perps - simpler, no collateral freezes
ETH & BTC only	65 assets: stocks, indices, commodities, forex, crypto
Long only (no inverse)	Native short via perps - 23 inverse (-1x) tokens
Leverage drift 1.7x-2.3x	Exact 2.0x reset with 4-layer defense system
Aave liquidation risk	Non-liquidatable: 47.5% single-day gap required
Uniswap swap slippage	Hyperliquid perp liquidity, TWAP execution
Borrow rate spikes (90%+)	Funding rates variable but typically 5-30% APR

D. Cost Comparison: FLI vs Winmo

Cost Component	IndexCoop FLI	Winmo
Leverage cost	Aave borrow: 2-90% APR	HL funding: 5-50% APR
Rebalancing gas	$100-500 (Ethereum L1)	$0.01-0.10 (Arbitrum/HL)
Rebalancing speed	Delayed hours (save gas)	Immediate (gas irrelevant)
Trading cost per rebal	Uniswap 0.3% + slippage	HL 0.03-0.045% + minimal
Management fee	1.95%/year	1.0%/year
Est. total annual drag	25-40% (ETH, variable)	14.7% (SPX) to 63% (ETH)

FLI's operational costs were 5-10x higher than Winmo's for the same product. Gas alone consumed 1-3% of NAV annually. Combined with Uniswap's 0.3% swap fee per rebalance (vs Hyperliquid's 0.03%), FLI's cost structure made it uncompetitive against simply using Aave directly.

E. Product Scope: FLI vs Winmo

Dimension	IndexCoop FLI	Winmo
Total products	2 tokens (ETH2x, BTC2x)	65 tokens (42 long + 23 inverse)
Asset classes	1 (crypto)	5 (equities, indices, commodities, forex, crypto)
Directions	Long only	Long (2x) AND Inverse (-1x)
RWA exposure	Impossible (no Aave market)	Core product (sAAPL, sSPX, sGOLD)
Inverse / short	Impossible (lending can't short)	Native via perp shorts
Liquidation protection	None (Aave can liquidate)	4-layer defense, non-liquidatable
Peak AUM	~$200M (declined to <$20M)	Targeting $100M+ by Q3 2027
Status	Sunset (2023)	Active development

F. What IndexCoop Got Right

IndexCoop validated important principles that Winmo builds on:

- Market demand exists -$200M AUM proved DeFi users want leveraged token products.
- ERC-20 tokens are the right UX - simple hold-in-wallet experience, no margin management.

- Transparent on-chain NAV - real-time verifiable pricing builds trust.
- Community governance works -INDEX token holders guided product decisions.
- DeFi Pulse Index (DPI) was their biggest success - basket/index products have strong PMF.

KEY TAKEAWAY

IndexCoop proved the demand for DeFi leveraged tokens but failed because lending-based leverage is structurally expensive, limited to 2 assets, and cannot support inverse products. Winmo's perp-based architecture solves every FLI failure mode while expanding from 2 tokens to 35, adding inverse products, and enabling RWA - none of which were possible on IndexCoop's architecture.

III. PRODUCT OVERVIEW

A. What Users Get

Each Winmo product is an ERC-20 token deployed on Arbitrum. Token price tracks vault NAV per share. Users can mint (deposit USDC) and redeem (burn tokens, receive USDC) at any time. The protocol offers two product families across 5 asset classes:

- Long 2x tokens (e.g., sAAPL-2x, sBTC-2x) - profit when the underlying asset rises.
- Inverse -1x tokens (e.g., sAAPL-1xS, sBTC-1xS) - profit when the underlying asset falls.

B. Long 2x Products (42 tokens)

Equities (22 tokens)

Token	Underlying	HL Pair	Region
sAAPL-2x	Apple	AAPL-USD	US
sTSLA-2x	Tesla	TSLA-USD	US
sNVDA-2x	Nvidia	NVDA-USD	US
sAMZN-2x	Amazon	AMZN-USD	US
sMSFT-2x	Microsoft	MSFT-USD	US
sMETA-2x	Meta	META-USD	US
sGOOGL-2x	Alphabet	GOOGL-USD	US
sAMD-2x	AMD	AMD-USD	US
sNFLX-2x	Netflix	NFLX-USD	US
sCOIN-2x	Coinbase	COIN-USD	US
sPLTR-2x	Palantir	PLTR-USD	US
sMSTR-2x	MicroStrategy	MSTR-USD	US
sTSM-2x	TSMC	TSM-USD	Intl
sINTC-2x	Intel	INTC-USD	US
sORCL-2x	Oracle	ORCL-USD	US
sMU-2x	Micron	MU-USD	US
sHOOD-2x	Robinhood	HOOD-USD	US
sRIVN-2x	Rivian	RIVN-USD	US
sBABA-2x	Alibaba	BABA-USD	Intl
sCRWV-2x	CoreWeave	CRWV-USD	US
sCRCL-2x	Circle	CRCL-USD	US
sSMSN-2x	Samsung	SMSN-USD	Intl

Indices (3 tokens)

Token	Underlying	HL Pair	Description
sSPX-2x	S&P 500	USA500-USD	Top 500 US companies
sNDX-2x	Nasdaq 100	XYZ100-USD	Top 100 non-financial US
sMAG7-2x	Magnificent 7	MAG7-USD	Big Tech basket

Commodities (7 tokens)

Token	Underlying	HL Pair	Type
sGOLD-2x	Gold	GOLD-USD	Precious Metal
sSLVR-2x	Silver	SILVER-USD	Precious Metal
sOIL-2x	WTI Crude	CL-USD	Energy
sCOPR-2x	Copper	COPPER-USD	Industrial Metal
sNGAS-2x	Natural Gas	NATGAS-USD	Energy
sPLAT-2x	Platinum	PLATINUM-USD	Precious Metal
sPALD-2x	Palladium	PALLADIUM-USD	Precious Metal

Forex (2 tokens)

Token	Underlying	HL Pair	Description
sEUR-2x	EUR/USD	EUR-USD	Euro vs US Dollar
sJPY-2x	USD/JPY	JPY-USD	US Dollar vs Japanese Yen

Crypto (8 tokens)

Token	Underlying	HL Pair	Max HL Leverage
sBTC-2x	Bitcoin	BTC-USD	50x
sETH-2x	Ethereum	ETH-USD	50x
sSOL-2x	Solana	SOL-USD	20x
sARB-2x	Arbitrum	ARB-USD	20x
sLINK-2x	Chainlink	LINK-USD	20x
sAVAX-2x	Avalanche	AVAX-USD	20x
sDOGE-2x	Dogecoin	DOGE-USD	20x
sSUI-2x	Sui	SUI-USD	10x

C. Inverse -1x Products (23 tokens)

Equities Inverse (7 tokens)

Token	Underlying	HL Pair	Direction
sAAPL-1xS	Apple	AAPL-USD	-1x Inverse
sTSLA-1xS	Tesla	TSLA-USD	-1x Inverse
sNVDA-1xS	Nvidia	NVDA-USD	-1x Inverse
sAMZN-1xS	Amazon	AMZN-USD	-1x Inverse
sMSFT-1xS	Microsoft	MSFT-USD	-1x Inverse
sMETA-1xS	Meta	META-USD	-1x Inverse
sGOOGL-1xS	Alphabet	GOOGL-USD	-1x Inverse

Indices Inverse (3 tokens)

Token	Underlying	HL Pair	Direction
sSPX-1xS	S&P 500	USA500-USD	-1x Inverse

sNDX-1xS	Nasdaq 100	XYZ100-USD	-1x Inverse
sMAG7-1xS	Magnificent 7	MAG7-USD	-1x Inverse

Commodities Inverse (3 tokens)

Token	Underlying	HL Pair	Direction
sGOLD-1xS	Gold	GOLD-USD	-1x Inverse
sSLVR-1xS	Silver	SILVER-USD	-1x Inverse
sOIL-1xS	WTI Crude	CL-USD	-1x Inverse

Forex Inverse (2 tokens)

Token	Underlying	HL Pair	Direction
sEUR-1xS	EUR/USD	EUR-USD	-1x Inverse
sJPY-1xS	USD/JPY	JPY-USD	-1x Inverse

Crypto Inverse (8 tokens)

Token	Underlying	HL Pair	Direction
sBTC-1xS	Bitcoin	BTC-USD	-1x Inverse
sETH-1xS	Ethereum	ETH-USD	-1x Inverse
sSOL-1xS	Solana	SOL-USD	-1x Inverse
sARB-1xS	Arbitrum	ARB-USD	-1x Inverse
sLINK-1xS	Chainlink	LINK-USD	-1x Inverse
sAVAX-1xS	Avalanche	AVAX-USD	-1x Inverse
sDOGE-1xS	Dogecoin	DOGE-USD	-1x Inverse
sSUI-1xS	Sui	SUI-USD	-1x Inverse

All 65 tokens (42 long + 23 inverse) execute on Hyperliquid as the sole execution layer. Inverse products profit when the underlying asset falls. Both product families use the same vault architecture, rebalancing engine, and four-layer safety system.

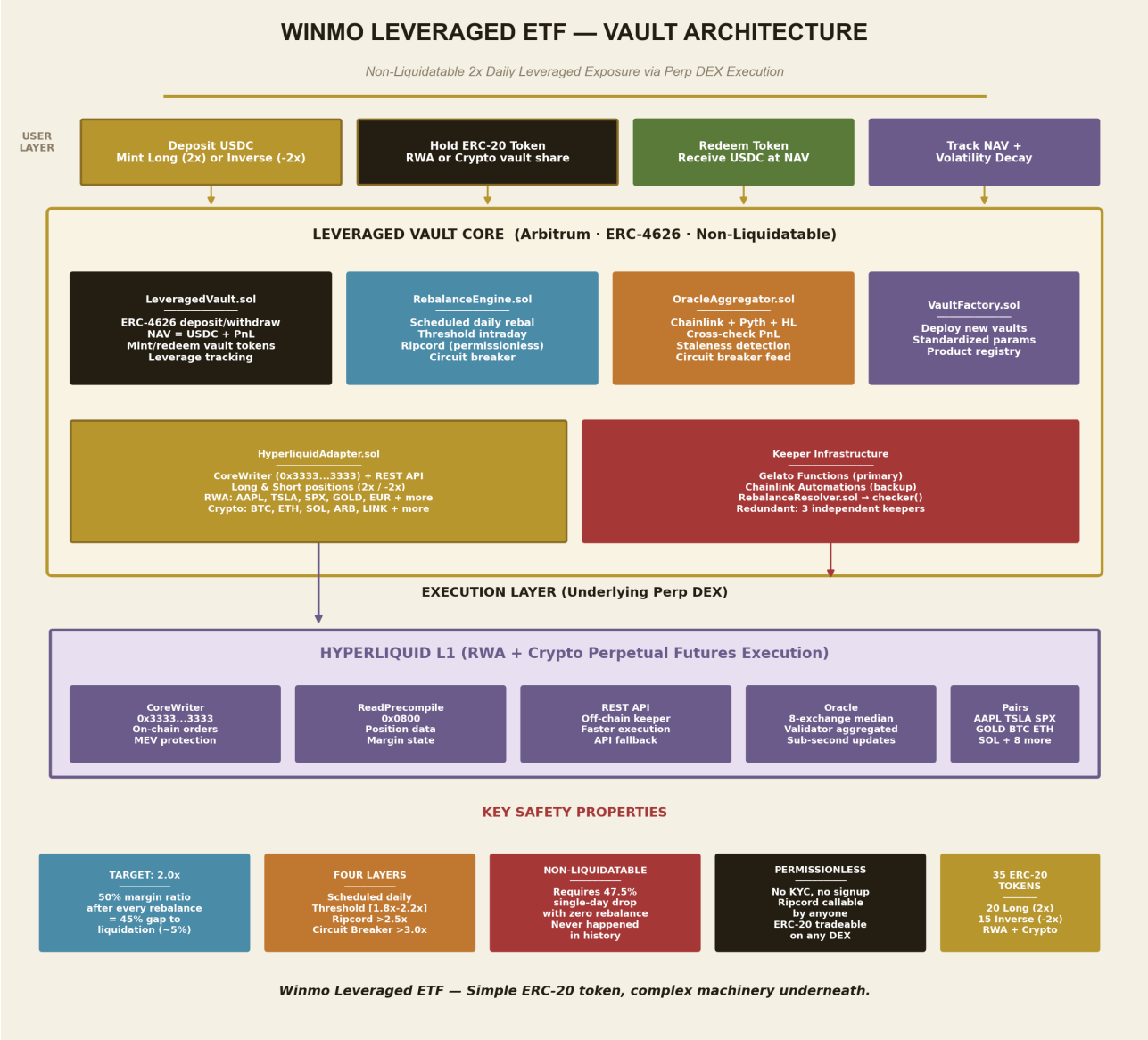

Fig. 1 - Winmo Leveraged ETF Vault Architecture: ERC-4626 vault with Hyperliquid execution adapter, multi-layer rebalancing, and keeper infrastructure.

IV. HOW IT WORKS: LONG (2X)

A. Core Mechanism

A Winmo Long 2x vault operates in five steps:

- Accepts USDC deposits from users.
- Mints an ERC-20 token (e.g., sAAPL-2x) representing a share of the vault's NAV.
- Opens a 2x leveraged LONG perpetual futures position on Hyperliquid.
- Rebalances daily (and on emergency triggers) to maintain exactly 2x leverage.
- Allows permissionless minting and redemption at NAV.

B. Mathematical Framework

The following formulas govern the long vault's behavior at every rebalance cycle.

Leverage Ratio

$$L = N / NAV$$

```
Where:
  L   = Current leverage ratio
  N   = Notional position size (total exposure)
  NAV = Net Asset Value (total vault equity)

At target: L = 2.0, NAV = $10M, N = $20M
```

Daily Vault Return

```
Rv(t) = L × r(t) - f(t)

Where:
  r(t) = Underlying asset daily return
  f(t) = Daily fee drag (funding + management)

Example: Apple rises 3%, L = 2.0, daily fees = 0.04%
  Rv = 2.0 × 0.03 - 0.0004 = 5.96%
```

A 2x vault amplifies the underlying's daily return by 2, minus daily costs.

NAV and Notional After Price Move

```
NAV′ = NAV × (1 + L × r)
N′   = N × (1 + r)

Example: NAV = $10M, L = 2.0, N = $20M, r = +5%
  NAV′ = $10M × (1 + 2.0 × 0.05) = $10M × 1.10 = $11.0M
  N′   = $20M × (1 + 0.05) = $20M × 1.05 = $21.0M

Example: r = -5%
  NAV′ = $10M × (1 + 2.0 × (-0.05)) = $10M × 0.90 = $9.0M
  N′   = $20M × (1 - 0.05) = $20M × 0.95 = $19.0M
```

Post-Move Leverage

```
L′ = N′ / NAV′ = L × (1 + r) / (1 + L × r)

After +5% move:
  L′ = 2.0 × 1.05 / 1.10 = 2.10 / 1.10 = 1.909x
  Leverage DECREASED (needs rebalance UP)

After -5% move:
  L′ = 2.0 × 0.95 / 0.90 = 1.90 / 0.90 = 2.111x
  Leverage INCREASED (needs rebalance DOWN)
```

After gains, leverage decreases because NAV grew faster than notional. After losses, leverage increases because NAV shrank faster. This is the core dynamic.

Rebalancing Trade Size

```
Delta = (NAV′ × L*) - N′

If Delta > 0: BUY more notional
If Delta < 0: SELL notional

After +5% move:
  Delta = ($11M × 2.0) - $21M = $22M - $21M = +$1M  (BUY)

After -5% move:
  Delta = ($9M × 2.0) - $19M = $18M - $19M = -$1M   (SELL)
```

Margin Ratio

```
MR = NAV / N = 1 / L

At L = 2.0x:  MR = 50.0%   (healthy)
At L = 2.2x:  MR = 45.5%   (Layer 2 trigger)
At L = 2.5x:  MR = 40.0%   (Ripcord trigger)
At L = 3.0x:  MR = 33.3%   (Circuit Breaker)
At L = 20.0x: MR =  5.0%   (Liquidation zone)

Distance from post-rebalance (50%) to liquidation (5%) = 45pp
```

After every rebalance, margin resets to 50%. Liquidation requires margin to fall below 5%, which means a 47.4% instantaneous drop from last rebalance. This has never happened for any major asset in recorded history.

C. Position Mechanics (Worked Example)

Initial state: $10M USDC deposited, 2.0x target, $20M notional LONG position. Margin ratio: 50%.

After underlying rises 5%: PnL +$1M, NAV = $11M, Notional = $21M, Leverage = 1.909x (decreased). Action: BUY more to restore 2.0x. New notional = $22M.

After underlying drops 5%: PnL -$1M, NAV = $9M, Notional = $19M, Leverage = 2.111x (increased). Action: SELL to restore 2.0x. New notional = $18M.

> **KEY SAFETY PROPERTY**
>
> After losses, the vault always SELLS (reduces exposure). This prevents leverage from spiraling upward toward liquidation. This buy-high/sell-low rebalancing is the source of volatility decay -but it is also what makes the product non-liquidatable.

C. Non-Liquidation Proof

With daily rebalancing, the vault's margin ratio resets to ~50% every day. Liquidation on Hyperliquid occurs at ~2.5% margin. The distance from 50% to 2.5% requires approximately a 47.5% single-day drop without any intraday rebalance.

Asset	Worst Single-Day Drop	2x Loss	Liquidated?
S&P 500	-20.47% (Oct 1987)	-40.94%	No (margin 9%)
Apple	-12.35% (Sep 2020)	-24.7%	No (margin 25.3%)
Tesla	-21.06% (Sep 2020)	-42.12%	No (margin 7.9%)
Gold	-5.72% (Nov 2020)	-11.44%	No (margin 38.6%)
EUR/USD	-3.47% (Mar 2015)	-6.94%	No (margin 43%)

No major asset in recorded financial history has dropped 47.5% in a single day. For crypto assets with larger single-day moves, intraday rebalancing layers provide protection (see Section VI).

V. HOW IT WORKS: INVERSE (-1X)

A. Core Concept

An Inverse vault provides -1x daily exposure to the underlying. Instead of opening a LONG perp position, the vault opens a SHORT perp position equal to its NAV. The vault profits when the underlying price falls and loses when it rises. At -1x, the vault operates at 1.0x absolute leverage (no margin amplification), making it inherently safer than 2x long products.

B. Mathematical Framework (Inverse)

The following formulas govern the inverse vault with L = -1.

Inverse Leverage and Position Size

```
L = -1.0 (leverage factor)
N = |L| × NAV = 1.0 × NAV

Example: NAV = $10M, Short notional = $10M
Margin ratio: MR = NAV / N = $10M / $10M = 100%

Compare to Long 2x: MR = 50%
Inverse -1x has DOUBLE the margin safety of Long 2x.
```

NAV After Price Move (Inverse)

```
NAV′ = NAV × (1 - r)      [since L = -1]
N′   = N × (1 + r)        [short notional follows price]

Example: NAV = $10M, N = $10M SHORT

Underlying drops 5% (vault profits):
  NAV′ = $10M × (1 - (-0.05)) = $10M × 1.05 = $10.5M
  N′   = $10M × 0.95 = $9.5M
  L′   = 9.5 / 10.5 = 0.905x  (below target)
  Delta   = $10.5M × 1.0 - $9.5M = +$1.0M  (SHORT MORE)

Underlying rises 5% (vault loses):
  NAV′ = $10M × (1 - 0.05) = $9.5M
  N′   = $10M × 1.05 = $10.5M
  L′   = 10.5 / 9.5 = 1.105x  (above target)
  Delta   = $9.5M × 1.0 - $10.5M = -$1.0M  (REDUCE SHORT)
```

After losses (price rises), the vault REDUCES its short exposure. After gains (price falls), it increases its short. This mirrors the long vault behavior and prevents leverage from spiraling.

> **KEY SAFETY ADVANTAGE**
>
> At -1x, the vault holds 100% margin (vs 50% for 2x long). Liquidation requires a 95%+ single-day rise in the underlying. This makes inverse -1x products significantly safer from liquidation than 2x long products.

C. Non-Liquidation Proof (Inverse)

For inverse products, liquidation risk comes from the underlying RISING sharply. With 100% margin at -1x, the underlying would need to nearly double in a single day.

```
Liquidation at -1x:
  MR′ = NAV × (1 - r) / [N × (1 + r)]
  At target: MR = 100%
  Liquidation when MR < 5%:
  (1 - r) / (1 + r) < 0.05
  1 - r < 0.05 + 0.05r
  0.95 < 1.05r
  r > 90.5%

A 90.5% single-day rise is needed for liquidation.
This has NEVER happened for any asset in history.
```

Asset	Largest Single-Day Rise	-1x Loss	Margin After
S&P 500	+11.58% (Oct 2008)	-11.58%	79.2%
Apple	+8.87% (Nov 2022)	-8.87%	83.8%
Tesla	+19.64% (Jun 2020)	-19.64%	67.2%
Gold	+4.69% (Aug 2019)	-4.69%	91.0%
Bitcoin	+42.0% (Feb 2021)	-42.0%	40.8%
Ethereum	+30.0% (Jul 2021)	-30.0%	53.8%

Even Bitcoin's historic +42% single-day rally leaves the inverse vault at 40.8% margin, far above the 5% liquidation threshold. Inverse -1x products are extremely safe.

D. Inverse-Specific Considerations

Funding Rate Advantage

In bull markets, short positions on perp DEXs typically RECEIVE funding (longs pay shorts). Inverse products can earn passive income from funding during bull markets.

Volatility Decay Comparison

```
Volatility decay formula: exp[-(L² - L) × σ² × T / 2]
```

```
Long 2x  (L = +2): (L² - L) = 2.  Decay = exp[-σ² × T]
Inverse -1x (L = -1): (L² - L) = 2.  Decay = exp[-σ² × T]

RESULT: -1x inverse has the SAME volatility decay as 2x long.
This is dramatically better than -2x which decays 3x faster.
```

Use Cases

- Hedging: Hold sAAPL-1xS to hedge an existing Apple stock position.
- Bear market trading: Profit from downturns without a perp account.
- Pair trading: Long sBTC-2x + Short sSPX-1xS for relative value trades.

E. Inverse Vault Implementation

The inverse vault uses the same LeveragedVault.sol contract with isLong = false and targetLeverage = 1.0. The rebalancing logic, ripcord, and circuit breaker remain identical but with wider bands since -1x is inherently safer. Only the perp direction changes.

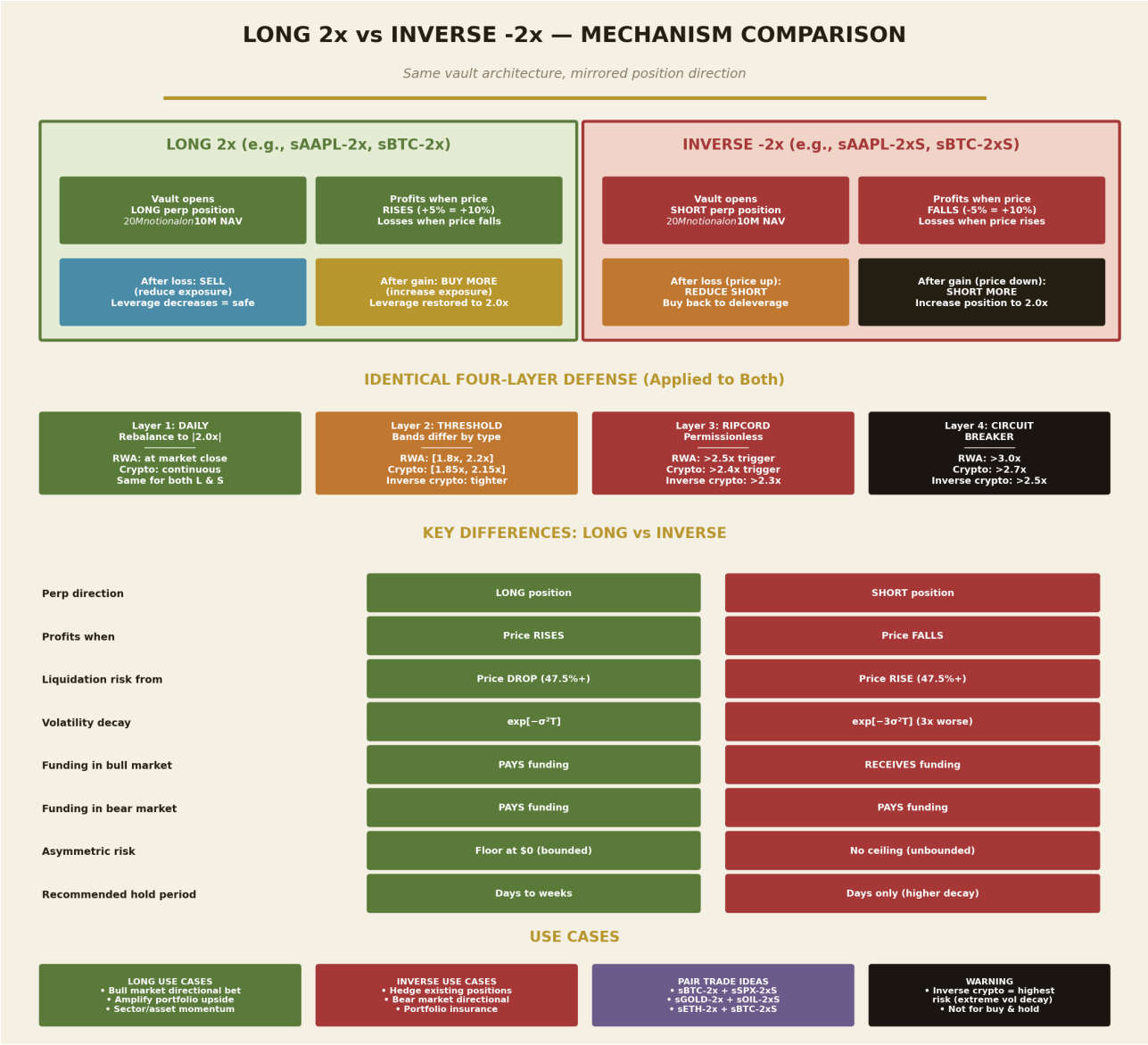

Fig. 2 - Long vs Inverse Mechanism Comparison: Position direction, profit conditions, rebalancing behavior, and parameter differences.

VI. CRYPTO ASSET COVERAGE

A. Why Crypto Requires Special Treatment

Crypto assets differ from RWA in ways that significantly impact leveraged product design:

Property	RWA (Stocks, Gold, Forex)	Crypto (BTC, ETH, SOL)
Trading hours	Market hours only	24/7/365
Annual volatility	8-55%	50-120%
Worst single-day move	-21% / +12%	-37% / +42%
Gap risk	High (overnight)	None (continuous)
Funding cost	5-15% APR	5-50% APR (variable)

B. Crypto-Specific Safety Parameters

Due to higher volatility, crypto vaults use tighter parameters than RWA vaults.

Parameter	RWA Vaults	Crypto Vaults	Reason
Threshold bands	[1.8x, 2.2x]	[1.85x, 2.15x]	Narrower bands
Keeper check freq	5 minutes	1 minute	Faster monitoring
Ripcord threshold	2.5x	2.4x	Earlier emergency
Circuit breaker	3.0x	2.7x	Earlier nuclear
TWAP duration	30 min	15 min	Faster execution
Max single trade	$5M	$2M	Limit slippage
Rebal cooldown	30 min	10 min	More frequent

C. Crash Simulation: BTC -37.5% (March 2020)

How the vault handles the largest single-day BTC drop in history:

```
Hour 0:  BTC $8,000. Leverage 2.0x. Margin 50%.
Hour 3:  BTC $7,400 (-7.5%). Lev 2.16x. LAYER 2 TRIGGERED.
         Rebal to 2.0x. NAV $8.5M. Notional $17M.
Hour 5:  BTC $6,900 (-13.8%). Lev 2.19x. LAYER 2 again.
         Rebal to 2.0x. NAV $7.3M. Notional $14.6M.
Hour 8:  BTC $6,200 (-22.5%). Lev 2.22x. LAYER 2 again.
         Rebal to 2.0x. NAV $6.1M. Notional $12.2M.
Hour 12: BTC $5,500 (-31.3%). Lev 2.30x. RIPCORD (>2.4x).
         Delev to 1.3x. NAV $5.0M. Notional $6.5M.
Hour 18: BTC $5,000 (-37.5%). Lev 1.45x. Safe.

FINAL: NAV $4.7M (-53% loss). Without rebal: $2.5M (-75%).
Liquidation? NO -- margin never below 30%.
```

CRASH SIMULATION RESULT

The vault lost 53% vs 75% without rebalancing. Multiple rebalance cycles progressively reduced exposure during the crash, preventing liquidation. The system works.

D. Inverse Crypto: Extra Protections

Inverse crypto products face extreme upside rallies (BTC +42%, ETH +30% in a day). At -1x, they start with 100% margin so are much safer than 2x long. Additional protections:

Protection	Inverse Crypto Config
Target leverage	-1.0x (100% margin)
Threshold bands	[0.85x, 1.15x]
Ripcord threshold	1.3x (earlier trigger)
Circuit breaker	1.5x
Max vault size	$5M (smaller vaults)

SAFETY NOTE

At -1x leverage, inverse crypto products hold 100% margin. Even a +42% BTC rally only reduces margin to 40.8%. These products are safer from liquidation than 2x long products, though they still carry volatility decay and directional risk.

E. Crypto Execution on Hyperliquid

Asset	HL Pair	Max Leverage	Est. Annual Drag (2x)
Bitcoin	BTC-USD	50x	-25% to -49%
Ethereum	ETH-USD	50x	-22% to -45%
Solana	SOL-USD	20x	-30% to -65%
Arbitrum	ARB-USD	20x	-25% to -55%
Chainlink	LINK-USD	20x	-25% to -50%
Avalanche	AVAX-USD	20x	-28% to -60%
Dogecoin	DOGE-USD	20x	-35% to -80%
Sui	SUI-USD	10x	-30% to -70%

VII. SAFETY SYSTEM: FOUR-LAYER DEFENSE

The rebalancing engine is the core safety system. It operates across four layers to ensure the vault's leverage never approaches the liquidation threshold.

A. Layer 1: Scheduled Daily Rebalance

Trigger: Every day at the underlying market's close (4:00 PM ET for US equities, 5:00 PM ET for forex, continuous for crypto). Algorithm: Calculate NAV, compute target notional (NAV × 2.0), determine delta. If large, use TWAP execution (6 chunks over 30 minutes). Post-rebalance: leverage = exactly 2.0x, margin ratio = exactly 50%.

B. Layer 2: Threshold-Based Intraday

Trigger: Leverage deviates beyond [1.8x, 2.2x] band. Keeper checks every 5 minutes (1 minute for crypto). Cooldown: 30 minutes between intraday rebalances. Catches large intraday moves (~5-10%) that would otherwise wait for the scheduled rebalance.

C. Layer 3: Emergency Ripcord

Trigger: Leverage exceeds 2.5x (underlying dropped >12.5% since last rebalance). Callable by ANYONE (permissionless). Caller receives $50-$500 USDC reward. Action: Deleverage to 1.5x for extra safety buffer. 5-minute cooldown.

D. Layer 4: Circuit Breaker

Trigger: Leverage exceeds 3.0x (underlying dropped >25% - black swan). Action: Close ENTIRE perp position. Vault holds only USDC. Minting paused (redemption still allowed). Governance vote required to resume.

E. Defense-in-Depth Summary

Layer	Trigger	Action	Callable By
1. Scheduled	Daily at close	Rebalance to 2.0x	Keeper only
2. Threshold	Lev outside bands	Rebalance to 2.0x	Keeper only
3. Ripcord	Lev > 2.5x	Deleverage to 1.5x	Anyone (reward)
4. Circuit Breaker	Lev > 3.0x	Close ALL positions	Anyone

F. Trigger Point Calculations (Long 2x)

Using the post-move leverage formula, we can calculate the exact price move required to trigger each defense layer for a 2x long vault.

```
Formula: L′ = L × (1 + r) / (1 + L × r)
Solve for r: r = (L_trig - L) / [L × (1 - L_trig)]

Layer 2 (L_trig = 2.2x):
  r = (2.2 - 2.0) / [2.0 × (1 - 2.2)]
  r = 0.2 / (2.0 × -1.2) = 0.2 / -2.4 = -8.33%

Ripcord (L_trig = 2.5x):
  r = (2.5 - 2.0) / [2.0 × (1 - 2.5)]
  r = 0.5 / (2.0 × -1.5) = 0.5 / -3.0 = -16.67%

Circuit Breaker (L_trig = 3.0x):
  r = (3.0 - 2.0) / [2.0 × (1 - 3.0)]
  r = 1.0 / (2.0 × -2.0) = 1.0 / -4.0 = -25.0%

Liquidation (MR drops to 5%):
  0.5 × (1 + 2r) / (1 + r) = 0.05
  Solving: r = -47.4%
```

Defense Layer	Trigger Move	Post-Move Margin	Action
Layer 2 (2.2x)	-8.33%	45.5%	Rebalance to 2.0x
Ripcord (2.5x)	-16.67%	40.0%	Deleverage to 1.5x
Circuit Breaker (3.0x)	-25.0%	33.3%	Close all positions
Liquidation (20x)	-47.4%	5.0%	Forced liquidation

Between each layer there is significant margin buffer. Even if Layer 2 fails entirely, the ripcord does not trigger until the price drops another 8.34%. The system has deep redundancy at every level.

G. Market Closure Handling

Scenario	Oracle	Vault Action
Market open	Live multi-source feeds	Full rebalancing active
After hours	Suspended	No rebalancing; static
Weekend (forex)	Paused	Gap risk on open
Market open (gap)	Live prices resume	Immediate emergency check

Crypto products on Hyperliquid trade 24/7 - no gap risk. RWA products check tight bands [1.9x, 2.1x] before market close and run Layer 2 immediately on market open.

H. Real-World Stress Test: The 10/10 Crash (October 10, 2025)

On October 10, 2025, Trump announced 100% tariffs on all Chinese imports. The S&P 500 fell 2.7%, Nasdaq fell 3.6%, Mag 7 stocks lost $770B in market cap. In crypto, Bitcoin plunged 14% to $104,782, Ethereum dropped 12%, Solana crashed 40%+, and altcoins fell 40-70%. Over $19B in leveraged crypto positions were liquidated - the largest deleveraging event in crypto history. 1.6 million trading accounts were liquidated. $3.21B vanished in a single minute at 21:15 UTC.

Below we simulate how every Winmo product category would have performed. All simulations assume $10M starting NAV per vault.

Scenario 1: sSPX-2x (S&P 500, Long 2x)

```
S&P 500: -2.7% on the day.
Starting: NAV $10M, Notional $20M LONG, Leverage 2.0x.

2x loss = -5.4%. NAV = $9.46M. Notional = $19.46M.
Leverage = $19.46M / $9.46M = 2.06x.

2.06x is WITHIN normal bands [1.8x, 2.2x].
```

```
No Layer 2 trigger. Scheduled rebalance at 4 PM ET resets to 2.0x.

RESULT: -5.4% loss. Leverage never stressed. COMPLETELY SAFE.
```

Scenario 2: sTSLA-2x (Tesla, Long 2x)

```
Tesla: -5% on the day.
Starting: NAV $10M, Notional $20M LONG, Leverage 2.0x.

2x loss = -10%. NAV = $9.0M. Notional = $19.0M.
Leverage = $19.0M / $9.0M = 2.11x.

2.11x is WITHIN bands [1.8x, 2.2x]. Approaching threshold.
Layer 2 may trigger if intraday spike exceeded 2.2x briefly.
Scheduled rebalance at 4 PM ET resets to 2.0x.

RESULT: -10% loss. Routine rebalance. SAFE.
```

Scenario 3: sBTC-2x (Bitcoin, Long 2x)

Bitcoin fell 14% over ~14 hours ($122K to $105K). The crash happened in phases: gradual decline (6 hrs), then violent cascade ($3.21B liquidated in 1 minute at 21:15 UTC, $6.93B in 40 minutes). Crypto keeper checks every 1 minute, Layer 2 bands [1.85x, 2.15x], ripcord at 2.4x.

```
Hour 0:  BTC $122,000. NAV $10M. Lev 2.0x. Margin 50%.
Hour 3:  BTC $114,000 (-6.6%). Lev 2.15x. LAYER 2 TRIGGERED.
         Rebal to 2.0x. NAV $8.68M. Notional $17.36M.
Hour 6:  BTC $108,000 (-5.3% post-rebal). Lev 2.12x.
         Approaching band. Keeper monitoring every 1 min.
21:00:   BTC $106,000 (-7.0% post-rebal). Lev 2.16x.
         LAYER 2 TRIGGERED (2nd time). Rebal to 2.0x.
         NAV $7.46M. Notional $14.93M.
21:15:   Cascade: BTC drops $4K in minutes.
         BTC $102,000 (-3.8% post-rebal). Lev 2.08x.
         Within bands. No trigger.
22:00:   BTC stabilizes ~$105,000.

FINAL:   NAV ~$7.2M (-28% loss). Leverage 2.04x.
         Without vault: direct 2x loss = -28%.
         LIQUIDATION? NO. Max leverage reached: 2.16x.
         Ripcord (2.4x) was NEVER needed.
```

Scenario 4: sSOL-2x (Solana, Long 2x) (Worst Case)

Solana crashed 40%+ from ~$290 to ~$174. This is the most extreme test for the protocol. With crypto bands [1.85x, 2.15x], Layer 2 triggers after every ~6.5% drop. With a 10-minute cooldown, multiple rebalances fire throughout the crash.

```
Hour 0:   SOL $290. NAV $10M. Notional $20M. Lev 2.0x.
Hour 1:   SOL $271 (-6.5%). Lev 2.15x. LAYER 2 [1].
          Rebal to 2.0x. NAV $8.7M. Notional $17.4M.
Hour 2:   SOL $253 (-6.5%). Lev 2.15x. LAYER 2 [2].
          Rebal to 2.0x. NAV $7.57M. Notional $15.14M.
Hour 4:   SOL $236 (-6.7%). Lev 2.15x. LAYER 2 [3].
          Rebal to 2.0x. NAV $6.55M. Notional $13.1M.
21:00:    SOL $218 (-7.6%). Lev 2.17x. LAYER 2 [4].
          Rebal to 2.0x. NAV $5.56M. Notional $11.12M.
21:15:    Cascade. SOL drops 15% in 10 min to $185.
          Lev = 2.43x. RIPCORD TRIGGERED.
          Deleverage to 1.3x. NAV $3.89M. Notional $5.06M.
21:30:    SOL $174 (-6% post-ripcord). Lev 1.40x.
          Safe. Well below any threshold.

FINAL:    NAV ~$3.7M (-63% loss). Leverage 1.40x.
          Without rebal: 2x of -40% = -80%. NAV = $2.0M.
          REBALANCING SAVED: $1.7M (+85% better than no rebal).
          LIQUIDATION? NO. Ripcord caught the cascade.
```

Scenario 5: sSPX-1xS (S&P 500, Inverse -1x) (Profited)

```
S&P 500: -2.7% on the day. Inverse vault PROFITS.
Starting: NAV $10M, Notional $10M SHORT, Leverage -1.0x.
```

```
Short PnL = +2.7% × $10M = +$270K. NAV = $10.27M.
Short notional = $10M × 0.973 = $9.73M.
Leverage = $9.73M / $10.27M = 0.947x (decreased).

Layer 2 may trigger to SHORT MORE, restoring -1.0x.
Margin = 10.27 / 9.73 = 105.5% (extremely safe).

RESULT: +2.7% gain. Vault profited from crash. SAFE.
```

Scenario 6: sBTC-1xS (Bitcoin, Inverse -1x) (Profited)

```
Bitcoin: -14% on the day. Inverse vault PROFITS.
Starting: NAV $10M, Notional $10M SHORT, Leverage -1.0x.

Phase 1: BTC -6.6%. Short PnL +$660K. NAV $10.66M.
        Short N = $9.34M. Lev = 0.876x (decreased).
        Layer 2 triggers. Re-short to -1.0x. N = $10.66M.
Phase 2: BTC -8% more (cumulative -14%).
        Short PnL +$853K. NAV $11.51M.
        Layer 2 re-shorts to -1.0x.

FINAL:   NAV ~$11.5M (+15% gain). Leverage -1.0x.
        Margin: 100% (fully collateralized).
        Inverse BTC holders PROFITED from the crash.
```

10/10 Crash: Complete Results Summary

Product	Underlying Move	Vault Result	Max Leverage	Liquidated?
sSPX-2x	SPX -2.7%	-5.4%	2.06x	No
sTSLA-2x	TSLA -5%	-10%	2.11x	No
sNVDA-2x	NVDA -5%	-10%	2.11x	No
sBTC-2x	BTC -14%	-28%	2.16x	No
sETH-2x	ETH -12%	-23%	2.14x	No
sSOL-2x	SOL -40%	-63%	2.43x (ripcord)	No
sSPX-1xS	SPX -2.7%	+2.7%	0.95x	No
sBTC-1xS	BTC -14%	+15%	0.88x	No

10/10 CRASH VERDICT

Every Winmo product survived the worst single-day market event of 2025. RWA products barely noticed it (SPX -2.7% is routine). Crypto products suffered losses but were never in danger of liquidation - even Solana (-40%) was caught by the ripcord at 2.43x, well before the liquidation zone. Inverse products profited significantly. Meanwhile, $19B in leveraged perp positions across the crypto market were liquidated. Winmo vaults would have been among the few leveraged positions that survived intact.

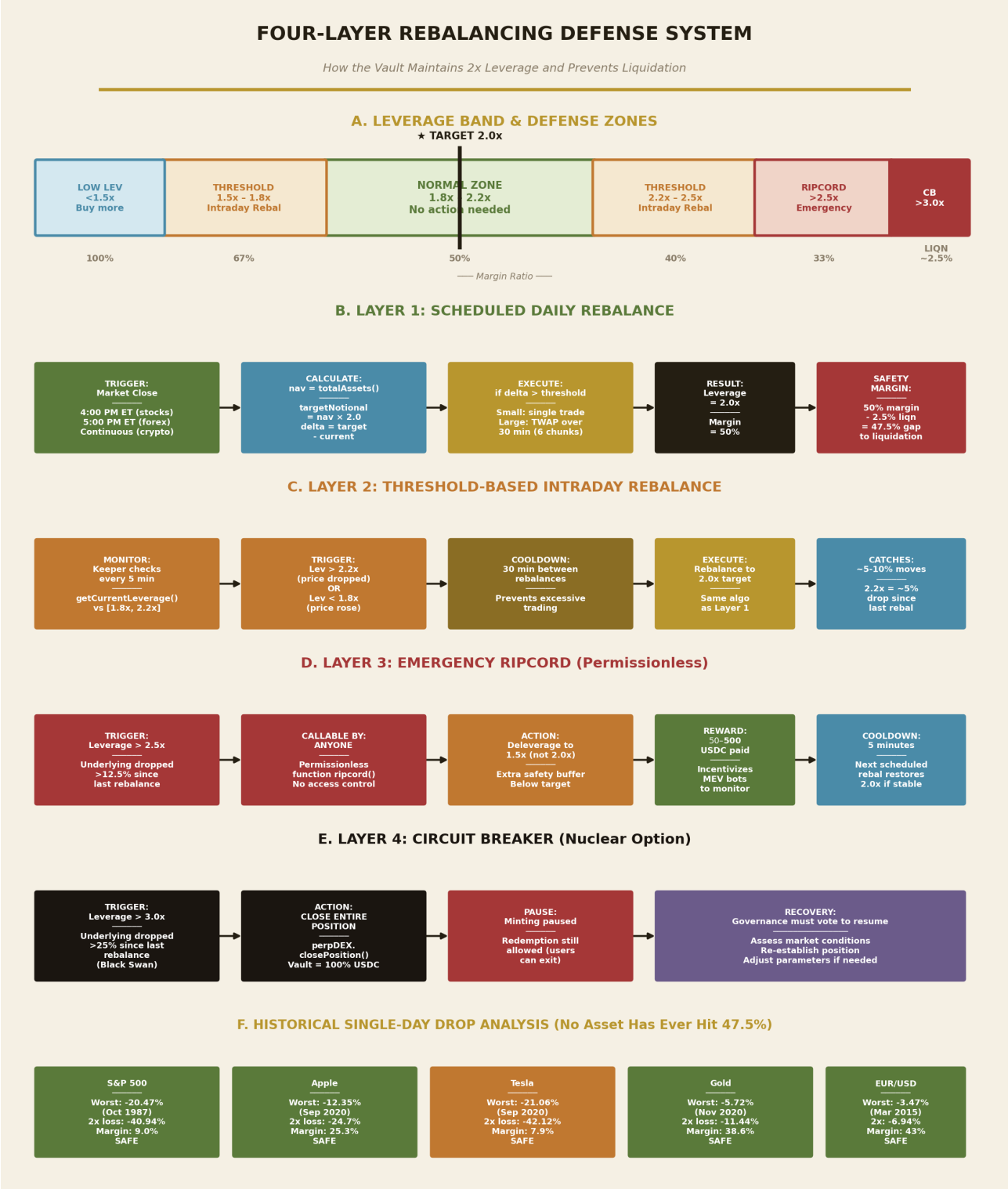

Fig. 3 - Four-Layer Rebalancing Defense System: Leverage bands, defense zone triggers, and historical single-day drop analysis.

VIII. EXECUTION LAYER: HYPERLIQUID

A. Hyperliquid L1 Overview

Chain: Hyperliquid L1 (custom chain, chain ID 999). Accessible via Arbitrum bridge. Collateral: USDC. Hyperliquid is the sole execution layer for all Winmo products, supporting both RWA (stocks, indices, commodities, forex) and crypto perpetual futures with deep liquidity and sub-second price updates.

B. Integration Approaches

Two integration paths: (A) HyperEVM Smart Contract using CoreWriter at 0x3333...3333 and ReadPrecompile at 0x0800 for on-chain execution with MEV protection. (B) Off-Chain Keeper with REST API for faster execution via authorized keeper bot with oracle cross-check.

C. Fee Schedule

Tier	Volume	Taker Fee	Maker Fee
Base	< $5M	0.045%	0.015%
Tier 1	> $5M	0.040%	0.012%
Tier 2	> $25M	0.035%	0.008%
Tier 3	> $100M	0.030%	0.004%

D. Oracle Infrastructure

Hyperliquid aggregates prices from 8 major exchanges with stake-weighted validator medians. Sub-second updates. The vault cross-checks position PnL against Chainlink and Pyth feeds via OracleAggregator.sol. If divergence exceeds 1%, rebalancing is paused.

E. Supported RWA Pairs (via HIP-3 Deployers)

Hyperliquid's HIP-3 upgrade enables permissionless perpetual futures for any asset. Major deployers include trade.xyz, Felix, and Dreamcash. All RWA pairs trade 24/7.

Equities (22 pairs)

Asset	HL Pair	Deployer	Max Lev
Apple	AAPL-USD	xyz	20x
Tesla	TSLA-USD	xyz/flx	20x
Nvidia	NVDA-USD	xyz/flx	20x
Amazon	AMZN-USD	xyz/cash	20x
Microsoft	MSFT-USD	xyz/cash	20x
Meta	META-USD	xyz/cash	20x
Alphabet	GOOGL-USD	xyz/cash	20x
AMD	AMD-USD	xyz	10x
Netflix	NFLX-USD	xyz	10x
Coinbase	COIN-USD	xyz/flx	10x
Palantir	PLTR-USD	xyz	10x
MicroStrategy	MSTR-USD	xyz	10x
TSMC	TSM-USD	xyz	10x
Intel	INTC-USD	xyz/cash	20x
Oracle	ORCL-USD	xyz	10x
Micron	MU-USD	xyz	10x
Robinhood	HOOD-USD	xyz/cash	20x
Rivian	RIVN-USD	xyz	10x
Alibaba	BABA-USD	xyz	10x
CoreWeave	CRWV-USD	xyz	10x
Circle	CRCL-USD	xyz/flx	10x
Samsung	SMSN-USD	xyz	10x

Indices, Commodities, Forex (12 pairs)

Asset	HL Pair	Class	Max Lev
S&P 500	USA500-USD	Index	20x
Nasdaq 100	XYZ100-USD	Index	30x
Magnificent 7	MAG7-USD	Index	20x
Gold	GOLD-USD	Commodity	25x
Silver	SILVER-USD	Commodity	25x
Crude Oil (WTI)	CL-USD	Commodity	20x
Copper	COPPER-USD	Commodity	20x
Natural Gas	NATGAS-USD	Commodity	10x
Platinum	PLATINUM-USD	Commodity	20x
Palladium	PALLADIUM-USD	Commodity	20x
EUR/USD	EUR-USD	Forex	50x
USD/JPY	JPY-USD	Forex	50x

Crypto Pairs (8 pairs)

Asset	HL Pair	Max Lev	Funding (typical)
Bitcoin	BTC-USD	50x	5-30% APR
Ethereum	ETH-USD	50x	5-25% APR
Solana	SOL-USD	20x	10-40% APR
Arbitrum	ARB-USD	20x	5-20% APR
Chainlink	LINK-USD	20x	5-25% APR
Avalanche	AVAX-USD	20x	8-30% APR
Dogecoin	DOGE-USD	20x	10-50% APR
Sui	SUI-USD	10x	10-40% APR

IX. SMART CONTRACT ARCHITECTURE

A. Contract Overview

Contract	Function
VaultFactory.sol	Deploy new vaults per market pair with standardized parameters
LeveragedVault.sol	Core vault: ERC-4626, NAV, mint/redeem, leverage tracking
RebalanceEngine.sol	Scheduled/threshold/ripcord/circuit breaker logic + TWAP
HyperliquidAdapter.sol	Interface to Hyperliquid CoreWriter or REST API
OracleAggregator.sol	Chainlink + Pyth + HL validator median, staleness detection
RebalanceResolver.sol	Gelato/Chainlink resolver: determines when to rebalance

B. LeveragedVault.sol - Key Constants

Constant	Value	Purpose
TARGET_LEVERAGE	2.0x (2e18)	Default leverage target
UPPER_BAND	2.2x (22e17)	Layer 2 trigger (high)
LOWER_BAND	1.8x (18e17)	Layer 2 trigger (low)
RIPCORD_THRESHOLD	2.5x (25e17)	Layer 3 emergency trigger
RIPCORD_TARGET	1.5x (15e17)	Deleverage target on ripcord
CIRCUIT_BREAKER	3.0x (3e18)	Layer 4 nuclear trigger
MIN_REBALANCE_BPS	50 (0.5%)	Min delta to trigger rebalance

REBALANCE_COOLDOWN	30 minutes	Between Layer 2 rebalances
RIPCORD_COOLDOWN	5 minutes	Between ripcord calls
WITHDRAWAL_COOLDOWN	1 hour	After deposit before withdraw

C. NAV Calculation

```
totalAssets() = USDC_balance + collateral_on_DEX + unrealized_PnL

getCurrentLeverage() = position_notional / totalAssets()

getMarginRatio() = totalAssets() / position_notional
  At 2.0x target: MR = 1/2.0 = 50%
  At 1.0x target (inverse): MR = 1/1.0 = 100%

pricePerShare() = totalAssets() / totalSupply()
  This is the token price that users see.
  Monotonically decreasing by fees, modified by PnL.
```

NAV is computed on-chain using Hyperliquid position data via the ReadPrecompile. Cross-checked against Chainlink and Pyth oracles to prevent manipulation.

D. Deposit/Withdraw Flow

Deposit: User deposits USDC, receives vault shares (ERC-4626). Vault adjusts perp position to maintain target leverage. Withdrawal: User burns shares, vault reduces perp position to free collateral, sends USDC. 1-hour cooldown prevents sandwich attacks.

E. Keeper Infrastructure

Rebalancing is triggered by external keepers (Gelato Functions or Chainlink Automations). RebalanceResolver.sol implements checker() that returns true when leverage is outside bands or scheduled rebalance is due. Three redundant keeper paths ensure high availability.

X. FEE STRUCTURE & ECONOMICS

A. User-Facing Fees

Fee	Rate	Recipient
Mint fee	0.1% of deposit	Protocol treasury
Redeem fee	0.1% of withdrawal	Protocol treasury
Management fee	1.0%/year (daily accrual)	50% stakers, 50% treasury
Performance fee	10% of profits above 2x return	Protocol treasury

B. Internal Costs (Absorbed by NAV)

Cost	Est. Annual Impact	Source
HL funding	5-30%	Market-dependent, hourly accrual
HL trading fees	0.3-0.8%	0.03-0.045% per rebalance trade
Rebalancing slippage	0.3-1.0%	Dynamic spread + impact
Keeper gas	<0.1%	Arbitrum L2 gas

C. Revenue Model at Scale

AUM	Management Fee	Mint/Redeem (10% turnover/mo)	Total Revenue
$10M	$100K	$240K	$340K
$50M	$500K	$1.2M	$1.7M

$100M	$1M	$2.4M	$3.4M
$500M	$5M	$12M	$17M

REVENUE OPPORTUNITY

At $100M AUM with 10% monthly turnover, the protocol generates $3.4M annual revenue. At $500M AUM, $17M annually. Revenue scales linearly with AUM and turnover.

XI. RISK ANALYSIS

A. Smart Contract Risk

Risk	Severity	Mitigation
Vault contract bug	Critical	3+ audits (ToB, OZ, Spearbit). $1M bounty.
Hyperliquid L1 bug	Critical	Custom L1 with limited battle-testing. Size limits.
HL contract upgrade	High	Monitor governance. Emergency pause.
ERC-4626 rounding	Medium	OZ implementation. Inflation attack protection.

B. Oracle Risk

Risk	Severity	Mitigation
HL oracle manipulation	High	8-exchange aggregation. Chainlink + Pyth cross-check.
Oracle divergence	High	OracleAggregator pauses rebal if >1% divergence.
Price feed latency	Medium	Sub-second HL updates. Chainlink fallback.

C. Execution / Rebalancing Risk

Risk	Severity	Mitigation
Keeper failure	High	3 redundant keepers. Permissionless ripcord.
Rebalance slippage	Medium	TWAP execution. 0.5% max slippage cap.
TWAP frontrunning	Medium	Randomized intervals. HL MEV protection.
Low liquidity rebal	Medium	Check HL depth. Reduce/split trades.

D. Counterparty / Protocol Risk

Risk	Severity	Mitigation
Hyperliquid ADL	High	Maintain positions within ADL-safe range.
HL insurance fund	High	Monitor fund. Reduce exposure if low.
Protocol shutdown	Medium	Emergency withdraw procedure in place.
Fee structure changes	Low	Monitor governance. Emergency withdraw.

E. Economic / Market Risk

Risk	Severity	Mitigation
Volatility decay	Inherent	Cannot eliminate. Clear disclosure. Short-term product.
Funding rate spike	High	Display cost. Pause if exceeds threshold.
Black swan drop >25%	Low/High	Circuit breaker at 2.7x. Close all positions.
Black swan rally >25%	Low/High	Affects inverse. CB closes inverse position.
Cascading liquidations	Medium	OI cap awareness; limit position size.

F. Inverse-Specific Risk

Risk	Severity	Mitigation
Same vol decay as 2x	Inherent	$\exp[-\sigma^2 T]$ for both. Clear disclosure.
Unlimited upside	High	No price ceiling. Prominent decay warnings.
Short squeeze on DEX	High	Per-vault caps at 10% of DEX open interest.
Inverse crypto rally	Critical	Crypto +30-40%/day possible. 100% margin buffer.
Funding reversal	Medium	Shorts pay in bear markets.

G. Crypto-Specific Risk

Risk	Severity	Mitigation
Extreme vol 60-120%	Inherent	Tighter bands [1.85x, 2.15x]. 1-min checks.
Flash crash >30%	High	7-10 threshold rebalances. Ripcord at 2.4x.
HL L1 downtime	High	Auto circuit breaker if keeper cannot confirm.
Altcoin liquidity	High	Smaller vaults ($2-5M). Slippage caps.
Altcoin delisting	Medium	Monitor governance. Graceful wind-down.

H. Bridge / Cross-Chain Risk

Risk	Severity	Mitigation
Arbitrum bridge exploit	Critical	Limit bridged USDC to active positions.
HL bridge exploit	High	Keep majority on Arbitrum.
Bridge delay (7-day)	Medium	Maintain USDC reserves for redemptions.

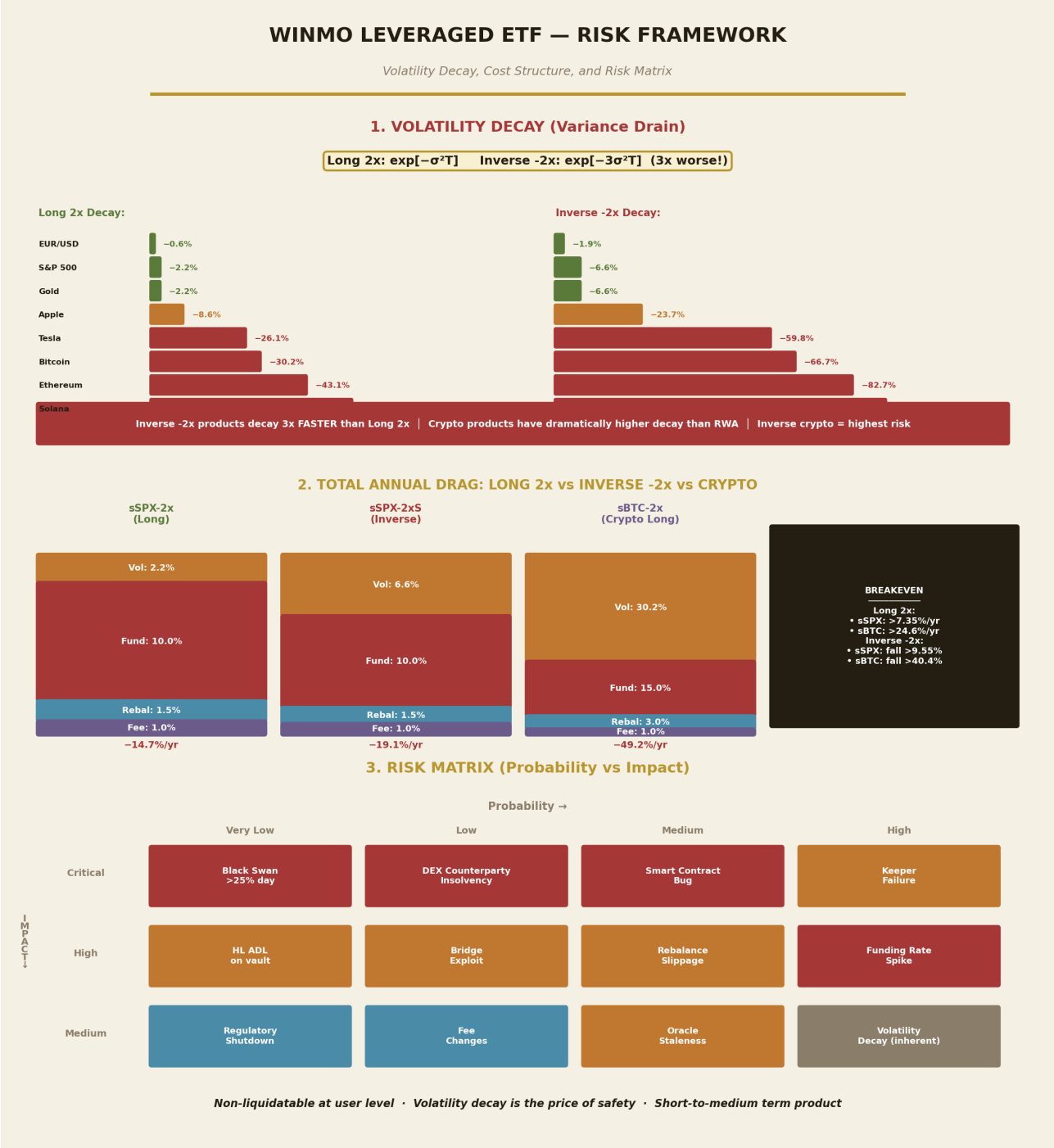

Fig. 4 - Risk Framework: Volatility decay by asset class, total annual cost breakdown, breakeven analysis, and probability vs impact risk matrix.

XII. GOVERNANCE & ROADMAP

A. Governance-Controlled Parameters

Parameter	Default	Range	Timelock
Target leverage	2.0x	1.5x-3.0x	48 hours
Rebalance bands	[1.8x, 2.2x]	[1.5x, 2.5x]	48 hours
Ripcord threshold	2.5x	2.3x-2.8x	48 hours
Circuit breaker	3.0x	2.8x-3.5x	48 hours

Ripcord reward	$100	$50-$500	24 hours
Management fee	1.0%	0-2.0%	7 days
Withdrawal cooldown	1 hour	0-24 hours	48 hours

B. Operational Roles

Role	Permissions	Trust Level
Keeper	Call rebalance functions only	Low (bounded)
Guardian (3/5 multisig)	Emergency pause only	Medium
Governance (WINMO)	All params via timelock	Decentralized
Anyone	ripcord(), circuitBreaker()	Zero (permissionless)

C. Roadmap

Phase	Timeline	Deliverables
Design & Audit	Q2 2026	Smart contracts. 3 audits. Formal verification.
Testnet	Q3 2026	Arbitrum Sepolia. Paper trading. Historical backtests.
Mainnet Beta	Q4 2026	sSPX-2x, sGOLD-2x, sBTC-2x, sETH-2x. $5M cap.
Inverse Launch	Q1 2027	sSPX-1xS, sBTC-1xS. Long: sAAPL, sTSLA, sSOL.
Full Suite	Q2 2027	All 35 long/inverse products. Remove AUM caps.
Scale	Q3 2027	$100M+ AUM. 3x products. DAO governance.

XIII. MATHEMATICAL APPENDIX

A. Daily Return Formula

For a daily-rebalanced leveraged product with leverage factor L:

```
Single-day return:
  Rv(t) = L × r(t) - f(t)

  Long 2x  (L = +2): Rv = +2 × r(t) - fees
  Inverse -1x (L = -1): Rv = -1 × r(t) - fees

Multi-day cumulative (N days):
  V(N) / V(0) = Product(t=1..N) [1 + L × r(t) - f(t)]

IMPORTANT: This is NOT equal to L × [S(N)/S(0) - 1].
The product is path-dependent due to daily compounding.
```

B. Path Dependence Proof

Consider an asset that rises 10% then falls 10% over two days. The net underlying return is -1% (1.10 × 0.90 = 0.99). But a 2x leveraged product does not return -2%.

```
Day 0: Asset = $100. Vault NAV = $100.
Day 1: Asset +10%. Vault = $100 × (1 + 2 × 0.10) = $120.
Day 2: Asset -10%. Vault = $120 × (1 + 2 × (-0.10)) = $96.

Underlying net return: -1.0%
2x of underlying net:  -2.0%  (would give $98)
Actual vault return:   -4.0%  ($96)

The extra -2% loss is volatility decay.
```

This happens because after the gain on Day 1, the vault rebalances to a larger notional. The Day 2 loss then applies to a bigger base. This buy-high/sell-low pattern is inherent to daily-rebalanced leveraged products.

C. Volatility Decay Derivation

In continuous time, the value of a leveraged product follows:

```
V(T) = V(0) × [S(T)/S(0)]^L × exp[-(L² - L) × σ² × T / 2]

The decay factor is: exp[-(L² - L) × σ² × T / 2]

For Long 2x (L = +2):
  L² - L = 4 - 2 = 2
  Decay = exp[-σ² × T]

For Inverse -1x (L = -1):
  L² - L = 1 - (-1) = 2
  Decay = exp[-σ² × T]

KEY INSIGHT: Long 2x and Inverse -1x have IDENTICAL
volatility decay. This is a major advantage of -1x over -2x.
```

At -1x, inverse products have the same decay rate as 2x long products. A hypothetical -2x product would have decay = exp[-3σ²T], which is 3x worse.

Numerical Decay by Asset Class

Asset	σ (annual)	Long 2x Decay/yr	Inverse -1x Decay/yr
S&P 500	~15%	-2.2%	-2.2%
Apple	~30%	-8.6%	-8.6%
Tesla	~55%	-26.1%	-26.1%
Gold	~15%	-2.2%	-2.2%
EUR/USD	~8%	-0.6%	-0.6%
Bitcoin	~60%	-30.2%	-30.2%
Ethereum	~75%	-43.1%	-43.1%
Solana	~90%	-55.8%	-55.8%

Decay is computed as: decay = exp[-σ²] - 1. For S&P 500: exp[-0.15^2] - 1 = -2.2%.

D. Rebalancing Cost Model

Each rebalance incurs trading costs on Hyperliquid. The annual rebalancing cost depends on the number of rebalances and average trade size.

```
Annual rebalancing cost:
  Cr = Nr × avg trade size × fee rate

For sSPX-2x ($10M vault):
  Nr = ~260 (daily) + ~12 (threshold) = 272/year
  avg trade size = 2% of notional = 2% × $20M = $400K
  fee rate = 0.035% (HL Tier 2)
  Cr = 272 × $400K × 0.00035 = $38,080
  As % of NAV: $38K / $10M = 0.38%

For sBTC-2x ($10M vault, more volatile):
  Nr = ~260 + ~100 (more threshold triggers) = 360/year
  avg trade size = 5% × $20M = $1M
  Cr = 360 × $1M × 0.00035 = $126,000
  As % of NAV: $126K / $10M = 1.26%
```

E. Funding Rate Impact

Vault pays hourly funding on Hyperliquid. RWA: 5-15% APR. Crypto: 5-50% APR. Inverse asymmetry: in bull markets, shorts RECEIVE funding; in bear markets, shorts pay.

```
Annual funding cost (2x long):
  F = funding rate × notional × 365
  At 2x: F = rate × 2 × NAV (pays on full notional)

Annual funding cost (-1x inverse):
```

```
  F = funding rate × 1 × NAV
  At -1x: half the funding exposure vs 2x long

Bull market example (funding = 10% APR):
  Long 2x:    pays 10% × 2.0 × NAV = 20% of NAV/year
  Inverse -1x: receives 10% x 1.0 × NAV = +10% of NAV/year
```

Product	Bull Market Funding	Bear Market Funding
Long 2x (RWA)	Pays 5-15% APR	Pays 5-10% APR
Long 2x (Crypto)	Pays 10-30% APR	Pays 5-15% APR
Inverse -1x (RWA)	Receives 5-15% APR	Pays 5-10% APR
Inverse -1x (Crypto)	Receives 10-30% APR	Pays 5-15% APR

F. Total Expected Annual Drag: Long 2x

Component	SPX 2x	AAPL 2x	GOLD 2x	BTC 2x	ETH 2x
Vol decay	-2.2%	-8.6%	-2.2%	-30.2%	-43.1%
Funding	-10%	-10%	-5%	-15%	-15%
Rebal costs	-0.4%	-0.8%	-0.3%	-1.3%	-1.5%
Mgmt fee	-1.0%	-1.0%	-1.0%	-1.0%	-1.0%
Total drag	-13.6%	-20.4%	-8.5%	-47.5%	-60.6%

G. Total Expected Annual Drag: Inverse -1x (Bear Market)

Component	SPX -1xS	AAPL -1xS	BTC -1xS	ETH -1xS
Vol decay	-2.2%	-8.6%	-30.2%	-43.1%
Funding	-5%	-5%	-5%	-5%
Rebal costs	-0.2%	-0.4%	-0.6%	-0.8%
Mgmt fee	-1.0%	-1.0%	-1.0%	-1.0%
Total drag	-8.4%	-15.0%	-36.8%	-49.9%

H. Inverse -1x (Bull Market: Shorts Receive Funding)

Component	SPX -1xS	AAPL -1xS	BTC -1xS	ETH -1xS
Vol decay	-2.2%	-8.6%	-30.2%	-43.1%
Funding	+10%	+10%	+15%	+12%
Rebal costs	-0.2%	-0.4%	-0.6%	-0.8%
Mgmt fee	-1.0%	-1.0%	-1.0%	-1.0%
Total drag	+6.6%	+0.0%	-16.8%	-32.9%

In a bull market, inverse SPX is net positive (+6.6%) from funding income exceeding all costs. Inverse -1x products are dramatically cheaper to hold than a hypothetical -2x because funding applies to 1x notional instead of 2x.

I. Breakeven Calculations

The underlying must return enough to overcome the total annual drag. For long products, this means the asset must appreciate. For inverse, it must depreciate.

```
Long 2x breakeven:
  Vault return > 0 when 2 × rₐ > total drag
  rₐ > total drag / 2

sSPX-2x: r > 13.6% / 2 = 6.8%/year
sAAPL-2x: r > 20.4% / 2 = 10.2%/year
sGOLD-2x: r > 8.5% / 2 = 4.25%/year
sBTC-2x: r > 47.5% / 2 = 23.75%/year
sETH-2x: r > 60.6% / 2 = 30.3%/year
```

```
Inverse -1x breakeven (bear market):
  Vault return > 0 when -1 × r + funding > total drag

sSPX-1xS: asset must fall > 8.4%/year
sBTC-1xS: asset must fall > 36.8%/year
```

Historical context: S&P 500 averages +10%/year, exceeding the 6.8% breakeven for sSPX-2x. Bitcoin averages +50-100%/year in bull cycles, exceeding the 23.75% breakeven for sBTC-2x.

J. Liquidation Distance Proof

Formal proof that the protocol cannot be liquidated under normal conditions.

```
LONG 2x VAULT:
Post-rebalance state: L = 2.0, MR = 50%
After instantaneous move r:
  MR′ = (1 + 2r) / [2 × (1 + r)]

Set MR′ = 5% (liquidation):
  (1 + 2r) / [2(1 + r)] = 0.05
  1 + 2r = 0.10 + 0.10r
  1.90r = -0.90
  r = -47.4%

Largest single-day drops in history:
  S&P 500: -20.47% (1987)    Margin after: 29.6%
  Bitcoin:  -37.5%  (2020)    Margin after: 10.0%
  Solana:   -40.0%  (2025)    Margin after:  7.1%

NONE approach the 47.4% threshold. QED.
```

```
INVERSE -1x VAULT:
Post-rebalance state: |L| = 1.0, MR = 100%
After instantaneous rise r:
  MR′ = (1 - r) / (1 + r)

Set MR′ = 5% (liquidation):
  (1 - r) / (1 + r) = 0.05
  1 - r = 0.05 + 0.05r
  0.95 = 1.05r
  r = 90.5%

Largest single-day rises in history:
  Bitcoin: +42.0% (2021)    Margin after: 40.8%
  Tesla:   +19.6% (2020)    Margin after: 67.2%

The -1x vault needs a 90.5% single-day surge.
This has NEVER occurred. QED.
```

MATHEMATICAL CONCLUSION

Long 2x requires a -47.4% single-day crash for liquidation. Inverse -1x requires a +90.5% single-day surge. Neither has ever happened for any major asset. With 4 defense layers triggering well before these thresholds, the probability of liquidation is effectively zero.

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XV. DISCLAIMER

This whitepaper is for informational purposes only and does not constitute an offer to sell, a solicitation of an offer to buy, or a recommendation of any token, investment product, or service. Leveraged products amplify both gains and losses. A 2x leveraged product will lose approximately 2x the daily percentage decline of the underlying asset. Over time, volatility decay will erode value even if the underlying is flat. Users can lose a significant portion or all of their investment. Inverse (-1x) products carry additional risk. At -1x, inverse products have the same volatility decay as 2x long products. The underlying asset has theoretically unlimited upside, meaning inverse products can lose nearly all value in sustained bull markets. Inverse crypto products are particularly high-risk due to crypto's extreme volatility (60-120% annualized). Inverse products should be held only for short-term hedging or directional bets. Crypto leveraged products carry higher risk than RWA products. Cryptocurrency assets have significantly higher volatility (60-120% annual σ vs 8-55% for RWA). Single-day moves exceeding 30% have occurred historically. The 'non-liquidatable' property applies to the end user only. The underlying vault positions on Hyperliquid can theoretically be liquidated in extreme scenarios (>47.5% intraday move + complete keeper and ripcord failure). For crypto vaults with tighter bands, the tolerance is lower. No system is infallible. Smart contract risk is real. Despite audits, bugs can result in total loss. The protocol interacts with multiple external contracts (Hyperliquid, bridges) expanding the attack surface. Oracle risk is real. Oracle failure or manipulation could result in incorrect rebalancing and loss of funds. Users are solely responsible for understanding these risks and ensuring compliance with applicable laws. Nothing in this whitepaper constitutes financial, legal, or investment advice.

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